Certified Collision-or-Separation Design for Finite RNA State Discrimination

Theoretical/computational RNA methods + statistical experimental design

D2T-RNA Project

Abstract

In a finite registered RNA state/action/observation model, the target and rival parameter classes can produce identical passive marginal laws, so deciding which class generated a fixed observed catalog requires a certified and checkable design. Classical marginal-fiber, Test-Cover, and fixed-horizon-testing arguments provide the component tools (fiber connectivity, test cover, sample-complexity scalings, costed design), but none by itself binds them into a single pre-registered finite RNA state/action/observation contract that emits a replayable collision-or-separation certificate over the complete cross-class difference set together with decision-theoretic and costed-design consequences. D2T-RNA is a decision-theoretic, auditable, and replayable certificate workflow: within a pre-registered finite RNA model it emits exact and independently checkable collision-or-separation certificates over the complete difference set, converts certified separation into fixed-horizon finite-sample decision and budget bounds that are *near-tight* against the exact minimax oracle, and returns either a costed integer design or a design-class no-go certificate used as a budget-constrained decision-support tool. All theoretical certificates are model-conditional and are cross-checked by rational primal/dual and independent-checker evidence against exact oracle enumeration over 11 synthetic micro-cases and 8 executed baselines; on a heterogeneous multi-pair instance the certified integer design is strictly cheaper than the greedy Test-Cover baseline (Remark 10). On a registered real RNA case (the add adenine-riboswitch aptamer, PDB 1Y26), the pipeline converts a previously fail-closed retrospective record into a registered observation-model instance and emits an exact separation certificate together with certified finite-sample and costed-design bounds, including a near-tight sample-complexity certificate and a no-go at unit budget; the remaining retrospective records (SAM-III, RORC) stay fail-closed. The measured certificate is reproduced across three real-data cases — the add riboswitch (1M7 SHAPE), the gcvT glycine riboswitch (DMS), and the miniTTR designed RNA switch (Mg^{2+}-dependent DMS) — so the registered channel is not riboswitch- or ligand-specific, and a real-data negative control confirms the executable is fail-closed. The contribution is a certified design method whose scope is bounded by the registered finite model.

1 Introduction

Field problem. Designing a probe panel to discriminate between two candidate models of a finite RNA system reduces, in the simplest setting, to deciding which of two parameter classes generated an observed catalog. When the passive observation is information-lossy, the two classes can be *observationally equivalent* under the marginal map, and no passive experiment can separate them.

What existing methods already solve. Markov-basis and marginal-fiber theory characterize when two contingency tables lie in the same fiber [Diaconis and Sturmfels, 1998, De Loera et al., 1995]; Test-Cover minimizes a set of tests that separate instances [Moret and Shapiro, 1991]; fixed-horizon active testing and sample-complexity analysis give scaling bounds for sequential discrimination [Chernoff, 1959, Hoeffding, 1963]; costed information-source selection and Bayesian experimental design optimize an allocation under cost or expected information [Pukelsheim, 2006, Lindley, 1956, Chaloner and Verdinelli, 1995]; active learning selects the most informative next probe [Cohn et al., 1996]; and RNA chemical-mapping heuristics (mutate-and-map, LM2R-style, SHAPE-directed modeling) [Kladwang et al., 2011, Cordero et al., 2012, Cheng et al., 2015, Hajdin et al., 2013] guide probe selection from experimental data.

Specific identifiability/design gap. These classical tools tell us *when* separation is possible in generic settings, but not how to produce a *replayable, checkable collision-or-separation certificate* over a specific finite registered model, and not what finite-sample decision/budget consequence follows from a certified separation. The complete cross-class difference set and the registered nuisance coupling are not articulated in the classical treatments.

Why generic cycle or Test-Cover arguments are insufficient. Generic alternating-cycle or Test-Cover arguments characterize fiber connectivity or test-cover existence, but they do not (i) pin the certificate to a complete cross-class difference set D rather than a hand-picked cycle subset, (ii) make the nuisance coupling explicit and pre-registered, or (iii) yield a directly auditable costed no-go certificate.

Research question. Within a finite registered model, can one certify collision-or-separation over the complete difference set, convert separation into fixed-horizon finite-sample decision/budget bounds, and return either a costed integer design or a design-class no-go certificate?

One-sentence core contribution. Within a pre-registered finite RNA state/action/observation model, D2T-RNA certifies exact collision-or-separation over the complete cross-class difference set, converts robust action-level separation into fixed-horizon finite-sample decision/budget bounds, and returns a costed integer design with a design-class no-go certificate, using rational primal/dual and independent-checker evidence.

Evidence and exact scope. The theorem certificates are model-conditional and synthetic, cross-checked by exact oracle enumeration over 11 micro-cases and 8 executed baselines. One real RNA case (the add adenine-riboswitch aptamer, PDB 1Y26) is additionally registered as a legitimate observation model — a binary protected/reactivity channel with a registered measurement-noise coupling — and yields an exact separation certificate with certified finite-sample and costed-design bounds (Section 8). The remaining public RNA records are used only as a fail-closed retrospective evidence audit within fixed realized datasets.

Claim boundary. We make no claim of universal RNA identifiability, population generalization, independent-library validation, prospective/blinded validation, or future wet-lab cost saving; the registered real-data case certifies separation *within its registered finite observation model*, not measured in-vitro reactivity. The contribution is a certified design method whose scope is the registered finite model.

2 Related work

2.1 Markov-basis and marginal-fiber identifiability

Diaconis–Sturmfels fiber theory and the Gröbner-basis treatment of the second hypersimplex describe when two tables lie in the same marginal fiber and which moves connect them [Diaconis and Sturmfels, 1998, De Loera et al., 1995]. We use these as building blocks to articulate the complete cross-class difference set; we do not claim a new connectivity theorem.

2.2 Bayesian experimental design and active learning

Bayesian experimental design optimizes an acquisition under a prior over parameters, from Lindley's information measure [Lindley, 1956] to the modern review of Chaloner and Verdinelli [1995]; active learning selects the most informative next point [Cohn et al., 1996]. These methods are *sequential and acquisition-based*: they return a policy or a point, not a finite-horizon certificate over a complete difference set. We inherit the information-theoretic objective (Hellinger/Bhattacharyya information) but fix a non-adaptive horizon and emit a replayable decision/budget certificate; we do not claim a new acquisition rule.

2.3 Controlled sensing and fixed-horizon testing

Fixed-horizon active hypothesis testing, sequential tests, and minimax sample-complexity results [Chernoff, 1959, Hoeffding, 1963, Wald, 1945, Lai and Robbins, 1985] give scaling bounds that we bind to a registered pair catalog and an explicit abstention rule, rather than repurposing them as a new sequential-testing theorem.

2.4 Costed experimental design

T-optimal and costed information-source selection [Pukelsheim, 2006] motivate our costed integer design. Our addition is an LP dual burden lower bound, an integrality gap, and a design-class no-go certificate over the registered catalog; the LP/test-cover formulation itself is inherited.

2.5 RNA chemical-mapping and retrospective evidence boundaries

SHAPE/DMS mutate-and-map and LM2R-style heuristics [Kladwang et al., 2011, Cordero et al., 2012, Cheng et al., 2015], SHAPE-directed modeling including pseudoknots [Hajdin et al., 2013], chemical-probing advances [Weeks, 2010], SHAPE-constrained folding [Sloma and Mathews, 2016], and the in-vitro vs. in-vivo probing gap [Leamy et al., 2016] guide probe selection from experimental data. Unregistered continuous channels are excluded from all categorical claims; public RNA data are confined to a fail-closed retrospective audit except where a legitimate registered binary channel is constructed (Section 8).

2.6 Where D2T-RNA sits in the design-testing lineage

Table 1 makes the four contrast dimensions explicit so the key difference is a checkable fact rather than a narrative. The fixed, non-adaptive panel is deliberate: it is what lets the certificate hold over the *complete* cross-class difference set D (Definition 2) rather than along one adaptive path, and what makes the certificate checkable by replay against a registered catalog (Section 8).

Table 1: How D2T-RNA differs from the design-testing lineage it builds on. The four columns are the dimensions a reader might conflate; only the last row is this paper's object.

	Bayesian experimental design	Active learning	Sequential testing	D2T-RN
Adaptivity	adaptive	adaptive	adaptive	*non-adaptive (fi*
Horizon	sequential	one-at-a-time	stopping time T	*fixed finit*
Certificate	acquisition value	next point	accept/reject + n	*collision-or-separation + finite-sample b*
Output	a design	a point/policy	a decision	*a replayable decision/b*

3 Registered finite RNA model

Definition 1 (Finite registered RNA model)**.** *A* finite registered RNA state/action/observation model *is a tuple* $(\Theta, A, \mathcal{Y}, M, \{B_u\}_{u\in A}, \mathcal{C}, h, \mathcal{R})$ *where:*

- $\Theta = \Theta_0 \sqcup \Theta_1$ *is a finite set of latent states, partitioned into a* target *class* Θ_0 *and a* rival *class* Θ_1*; the two classes are* disjoint *(*$\Theta_0 \cap \Theta_1 = \emptyset$*), so no pair in Definition 2 is the trivial diagonal pair* $\theta_0 = \theta_1$*;*
- A *is a finite set of* actions *(probes/perturbations);*
- $\mathcal{Y}$ *is a finite categorical* observation alphabet*;*
- $M : \Theta \to \Delta(\mathcal{Y})$ *is the* passive marginal map*;*
- *for each* $u \in A$*,* $B_u : \Theta \to \Delta(\mathcal{Y})$ *is the* action-level observation law*;*
- $\mathcal{C}$ *is a registered* nuisance coupling *(CARTESIAN or EQUAL_REALIZED_VALUE) that authorizes the action-level product law;*
- $h \in \mathbb{Z}_{\geq 0}$ *is a fixed* non-adaptive horizon*;*
- $\mathcal{R}$ *is an* abstention rule *(decision-or-abstain).*

Definition 2 (Complete cross-class difference set D)**.** *The* complete cross-class difference set *is* $D = \Theta_0 \times \Theta_1$*, the set of all target/rival pairs. It is the full cross product, not a hand-picked cycle subset. Because* Θ_0 *and* Θ_1 *are finite and disjoint (Definition 1),* D *is finite and contains no trivial diagonal pair* (θ, θ)*.*

Remark 1 (Relation to the contract difference set)**.** *This paper's* $D = \Theta_0 \times \Theta_1$ *is the finite-catalog representation of the contract's cross-class difference set: each pair* $(\theta_0, \theta_1) \in D$ *indexes the difference of the two latent state distributions, and certifying separation over the* full *cross product (rather than only the passive collision fiber, i.e. those pairs whose passive marginal laws coincide) is a conservative, stronger certificate. It is the full cross product, not a hand-picked subset.*

Remark 2 (Finite catalog, attainment, and triviality)**.** *The paper restricts the model to* finite *state classes (finite catalog), so* D *is finite and the minimum in Definition 4 is attained. Consequently* $\gamma(S) = 0$ *always yields a* non-trivial, attained *exact collision witness* $(\theta_0, \theta_1) \in D$ *with* $\theta_0 \neq \theta_1$ *(disjointness excludes the diagonal). The contract's general framework also allows* rational polyhedral *classes, in which case* D *is a difference set of state distributions and the infimum may not be attained; there the equivalence degrades to necessary-only / sufficient-only forms and the infimum-only obstruction boundary (contract 5.2), which is out of scope here. We therefore do* not *claim an attained witness when the infimum is only approached.*

Definition 3 (Action panel and induced law)**.** *For a panel* $S \subseteq A$ *of actions, the* induced observation law *of state* θ *is the product law* $P_S(\theta) = \bigotimes_{u\in S} B_u(\theta)$*, where the product is authorized by the registered coupling* $\mathcal{C}$*.*

Remark 3 (Passive marginal, induced law, and the action map). *M is the* passive *(no-action) observation law and defines the collision fiber: a pair (θ_0, θ_1) is passively collision-equivalent exactly when the two induced passive laws coincide. The action map B_u induces, on the categorical alphabet $\mathcal{Y}$, the atomic observation law used to* separate *the pair; the complete* categorical observation law *for a panel S is the product law $P_S(\theta) = \bigotimes_{u \in S} B_u(\theta)$ over $\mathcal{Y}^{|S|}$, authorized by the registered coupling $\mathcal{C}$. Thus $\gamma(S)$ measures, in* observation-probability *terms, how well the action maps separate every target/rival pair in D.*

Definition 4 (Separation functional $\gamma(S)$)**.** *$\gamma(S) = \min_{(\theta_0,\theta_1)\in D} \mathrm{TV}\big(P_S(\theta_0), P_S(\theta_1)\big)$, the minimum total-variation separation of the induced laws over the complete difference set. Explicitly, $\gamma(S)$ is a norm on pairs of* categorical observation-probability *laws over $\mathcal{Y}^{|S|}$ (the product law P_S of Definition 3),* not *a metric on the state space Θ. Because D is finite and the laws are finite categorical laws, the minimum is attained (Remark 2).*

Assumption 1 (Registered structure)**.** *The model is pre-registered: the state classes, action set, alphabet, marginal and action maps, coupling $\mathcal{C}$, horizon h, and abstention rule $\mathcal{R}$ are fixed before any data are used.*

Assumption 2 (Complete action-level likelihoods)**.** *For every pair in D and every action $u \in A$, the action-level likelihoods $B_u(\theta_0)(y)$ and $B_u(\theta_1)(y)$ are known and positive, so the product law is fully specified.*

Explicitly not covered. Continuous infinite-dimensional nuisance, adaptive actions, randomized design, sequential stopping, arbitrary high-dimensional contingency tables, unregistered continuous SHAPE/DMS channels, unknown action-induced state change, and unmodeled third states.

4 Exact geometry (T2b)

Theorem 1 (Exact collision-or-separation, T2b)**.** *Let the model of Definition 1 satisfy Assumptions 1–2. For a panel $S \subseteq A$:*

1. $\gamma(S) = 0$ if and only if *there exists $(\theta_0, \theta_1) \in D$ such that $P_S(\theta_0) = P_S(\theta_1)$ (an* exact collision witness*);*
2. $\gamma(S) > 0$ if and only if *$P_S(\theta_0) \neq P_S(\theta_1)$ for every $(\theta_0, \theta_1) \in D$ (a* separation certificate*).*

Moreover, when $\gamma(S) > 0$, the pair attaining the minimum yields a rational primal/dual certificate and an independent-checker verification.

Proof sketch. For finite D, the map $(\theta_0, \theta_1) \mapsto \mathrm{TV}(P_S(\theta_0), P_S(\theta_1))$ takes finitely many values, so its minimum is attained (Definition 4). If the attained minimum is 0, some pair has TV $= 0$, i.e. identical product laws, an exact collision witness, giving the forward direction of (i); the reverse direction is immediate. If the minimum is > 0, every pair has positive TV, i.e. distinct product laws, a separation certificate; the reverse direction is immediate. The rationality follows because all laws are finite rational categorical laws and the certificate is witnessed by an explicit pair; an independent checker re-derives the collision or separation from first principles (`verify.py verify_collision/verify_separation`) and the LP strong-duality + primal feasibility check supplies the rational primal/dual pair (`lp.py strong-duality`). The full proof is in the supplementary. □

Remark 4 (Direction semantics)**.** *For finite catalogs the infimum in $\gamma(S)$ is attained, so the theorem is* iff*. Both registered couplings ($\mathcal{C}$ = CARTESIAN and $\mathcal{C}$ = EQUAL_REALIZED_VALUE)* *fully specify the product law under Assumption 2, so the iff holds under either coupling. The necessary-only / sufficient-only forms are reserved for the non-atomic / infimum-only obstruction boundary (contract 5.2), which we do not treat here (Remark 2).*

Remark 5 (What the rational certificate verifies)**.** *The certificate is a statement about* observation-probability separation*, not a purely geometric* (Θ) *claim. The attaining pair* $(\theta_0^\star, \theta_1^\star) \in D$ *with* $\mathrm{TV}(P_S(\theta_0^\star), P_S(\theta_1^\star)) = \gamma(S)$ *is a witnessed element of* D*, and the value* $\gamma(S) > 0$ *is the categorical-law separation it certifies. The LP dual is an* intermediate *certificate for the costed-design consequence (T2d): it certifies that every feasible design has cost at least the dual bound, not a second, independent geometric claim. The independent checker re-derives both the witnessed collision/separation and the dual from first principles (`verify.py` and `lp.py`).*

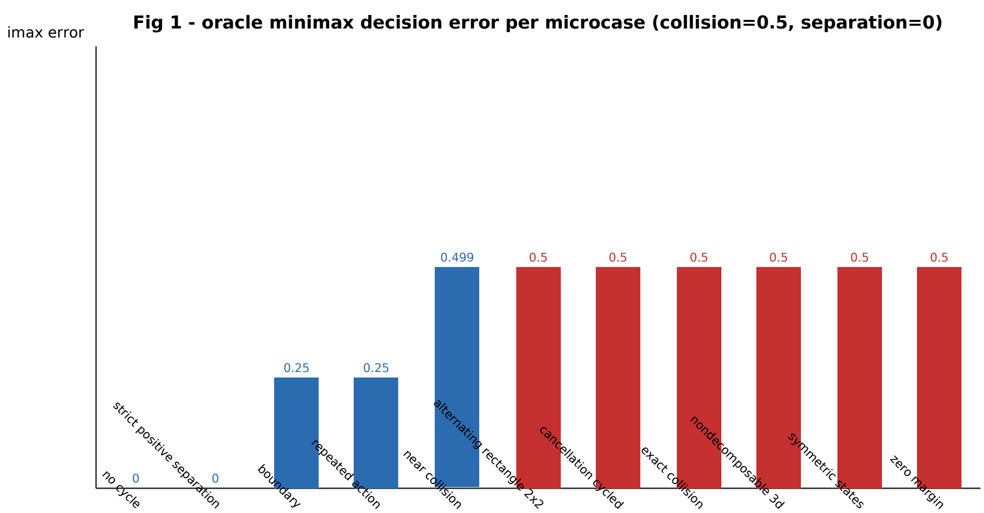

Figure 1: Exact geometry (T2b). **Question:** does the induced law map separate the complete difference set D? **Result:** oracle minimax decision error per synthetic microcase (model-conditional); colliding pairs reach error 1/2, separated pairs reach error 0. **Interpretation:** exact collision-or-separation certificate consistent with the oracle. **Boundary:** model-conditional; no population generalization.

5 Finite-sample consequence (T2c)

Definition 5 (Pair information)**.** *For a pair* $(\theta_0, \theta_1) \in D$ *and action* u*, the* Hellinger/Bhattacharyya information *[Hellinger, 1909] is* $I_u(\theta_0, \theta_1) = -\log \sum_{y \in \mathcal{Y}} \sqrt{B_u(\theta_0)(y)\, B_u(\theta_1)(y)}$*. Under* n_u *repeated non-adaptive observations of action* u *with the registered product law, the total information is* $I_{\mathrm{tot}} = \sum_{u \in A} n_u\, I_u(\theta_0, \theta_1)$*.*

Definition 6 (Decision-theoretic endpoints)**.** *Fix a* decision-or-abstain *rule* $\mathcal{R}$ *that, on each realized catalog, returns either a declared class or* inconclusive *(abstain). The registered endpoints are:*

- α*: the type-I wrong-declaration bound (declaring the rival when the target generated the data);*

- *β: the type-II wrong-declaration bound (declaring the target when the rival generated the data);*
- *κ_0: the minimum correct-declaration probability under the target hypothesis (H0);*
- *κ_1: the minimum correct-declaration probability under the rival hypothesis (H1);*
- *$\rho = \mathbb{P}(abstain)$: the probability the rule returns* inconclusive.

Every declaration is exactly one of correct, wrong, or abstained; abstention is neither correct nor wrong, so $\mathbb{P}(\text{correct}) + \mathbb{P}(\text{wrong}) + \rho = 1$.

Theorem 2 (Finite-sample decision bound, T2c)**.** *Consider a finite registered pair catalog under a fixed-horizon deterministic non-adaptive allocation with complete action-level likelihoods and registered product laws, with endpoints as in Definition 6.*

1. (Upper bound.) *For every pair, the probability of a wrong declaration satisfies $\mathbb{P}(\text{wrong}) \le \frac{1}{2}\exp(-I_{\text{tot}})$, so in particular $\alpha, \beta \le \frac{1}{2}\exp(-I_{\text{tot}})$; the correct-declaration probability satisfies $\mathbb{P}(\text{correct}) \ge 1 - \frac{1}{2}\exp(-I_{\text{tot}}) - \rho$, where ρ is the abstention probability under rule $\mathcal{R}$.*
2. (Lower bound / structural no-go.) *To reach a correct-declaration floor $\kappa = \max(\kappa_0, \kappa_1) \in (1/2, 1)$, the total information must satisfy $I_{\text{tot}} \ge -\frac{1}{2}\log\bigl(1 - (2\kappa - 1)^2\bigr)$. If the certified total information is strictly below this threshold, no allocation within the budget can meet the floor.*

Proof sketch. For (i), the Bhattacharyya bound gives $\mathbb{P}(\text{wrong}) \le \frac{1}{2}\prod_{u\in A}\rho_u^{n_u}$, where $\rho_u = \sum_y \sqrt{B_u(\theta_0)(y)B_u(\theta_1)(y$ e^{-I_u}, so the product is $e^{-I_{\text{tot}}}$. For (ii), the decision error is minimized when the two product laws are separated by the largest achievable TV; relating the required separation to the Bhattacharyya coefficient and rearranging the floor condition $\mathbb{P}(\text{correct}) \ge \kappa$ yields the stated information threshold. The arithmetic uses exact rationals (`fractions.Fraction`) and directed-rounding decimal intervals (`Decimal`) for the transcendental terms; the upper bound is cross-checked against exact equal-prior Bayes-average enumeration (`decision.py exact_bayes_average_error`). The full proof is in the supplementary. □

Remark 6 (Tightness microcase)**.** *For alphabet size 2, $P_0 = (1/4, 3/4)$, $P_1 = (1, 0)$: the Bhattacharyya coefficient is $1/2$, the information is $\ln 2$, the exact equal-prior Bayes-average error is $\frac{1}{2}(1/4)^n$, and the certified upper bound $\frac{1}{2}\exp(-nI)$ brackets the Bhattacharyya-ideal $\frac{1}{2}(1/2)^n$, i.e. the finite-sample bound dominates the exact equal-prior Bayes-average error and is conservative by the intended margin.*

Remark 7 (Budget and sample complexity)**.** *To meet a correct-declaration floor $\kappa = \max(\kappa_0, \kappa_1)$ with abstention ρ, T2c (ii) requires $I_{\text{tot}} \ge -\frac{1}{2}\log\bigl(1-(2\kappa-1)^2\bigr)$. Expressed as a budget, the minimal non-adaptive allocation is the integer solution of $\min_{n\in\mathbb{Z}^A_{\ge 0}}\sum_{u\in A} c_u n_u$ subject to $\sum_{u\in A} n_u I_{uw} \ge -\frac{1}{2}\log\bigl(1-(2\kappa_w-1)^2\bigr)$ for every active pair w, which is exactly the T2d program (Section 6) with τ_w set to that per-pair threshold. Thus T2c supplies the per-pair information/budget target and T2d turns it into a costed design or a design-class no-go. The bounds are stated for the finite registered pair catalog only; no composite-continuous covering or real-data calibration guarantee is made.*

6 Costed design and no-go (T2d)

Definition 7 (Costed design problem)**.** *Given unit costs $c_u \ge 0$ and requirement thresholds $\tau_w \ge 0$ over active pairs $w \in D'$, the* costed design problem *is*

$$\min_{n\in\mathbb{Z}^A_{\ge 0}} \sum_{u\in A} c_u n_u \quad s.t. \quad \sum_{u\in A} n_u I_{uw} \ge \tau_w \quad \forall w \in D',$$

where I_{uw} is the certified pair-information interval for pair w under action u.

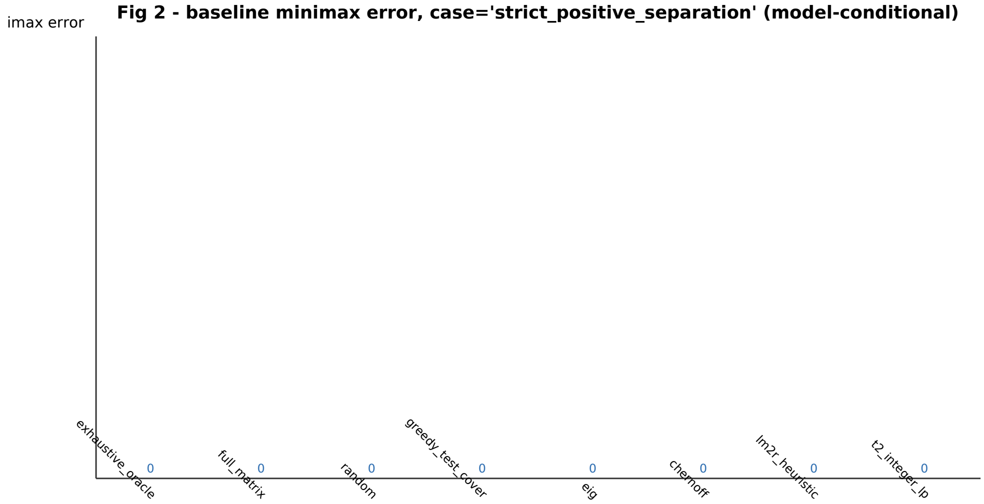

Figure 2: Finite-sample consequence (T2c). **Question:** how does the certified decision bound compare with exact oracle enumeration? **Result:** baseline minimax decision error by exact oracle enumeration for a representative separated case (model-conditional). **Interpretation:** the finite-sample decision consequence cross-checks the exact enumeration. **Boundary:** no composite-continuous covering theorem, no arbitrary continuous nuisance guarantee, no real-data calibration.

Definition 8 (LP relaxation and dual)**.** *The* LP relaxation *relaxes* $n_u \geq 0$ *to real. Its* dual *is* $\max_{y\geq 0} \sum_{w\in D'} \tau_w y_w$ *subject to* $\sum_{w\in D'} I_{uw}\, y_w \leq c_u$ *for all* $u \in A$*. For any dual-feasible* y*,* $\sum_w \tau_w y_w$ *is a* dual burden lower bound *on the cost of every feasible design.*

Theorem 3 (Costed design and no-go, T2d)**.** *Let the model satisfy Assumptions 1–2.*

1. (Achievable upper.) *If an integer point* $n^\star$ *satisfies* $\sum_u n^\star_u I^{\mathrm{lo}}_{uw} \geq \tau_w$ *for all* $w \in D'$*, then* $c^T n^\star$ *is an achievable cost (upper bound) and its feasibility is explicitly checkable.*
2. (No-go lower.) *If a dual-feasible* y *gives* $\tau^T y >$ budget*, then no feasible design within budget exists (a* design-class no-go certificate*).*
3. (Integrality gap.) gap $= \big($*upper cost* $-$ *lower bound*$\big)/$*lower bound, computed conservatively with certified interval information.*

Proof sketch. (i) is by construction: any integer $n^\star$ satisfying the coverage constraints is feasible and has cost $c^T n^\star$; feasibility is re-checked from first principles (`costed_verify.check_integer_design_feasible`). (ii) follows by weak duality: every feasible design satisfies $\sum_u c_u n_u \geq \sum_w \tau_w y_w$ for every dual-feasible y (using the interval upper bound on I_{uw} keeps the LP a relaxation), so a dual bound exceeding the budget rules out all feasible designs. (iii) is direct arithmetic. The full proof is in the supplementary. □

Remark 8 (Exact microcases)**.** Single-pair integral: $\min n$ *with* $n \geq 3$*, cost* 2 *gives IP=LP=* 6*, gap* 0. Fractional cover 3×3: $A = [[1,0,1],[1,1,0],[0,1,1]]$*,* $\tau = (1,1,1)$*,* $c = (1,1,1)$ *gives LP=* $3/2$*, IP=* 2*, gap* $= 1/3$. Fractional requirement: $\min n$ *with* $n \geq 3/2$*, cost* 1 *gives LP=* $3/2$*, IP=* 2*, gap* $= 1/3$. No-go demonstration: *a fractional cover with budget* $7/5 < 3/2 =$ *LP bound is NO_GO.*

Remark 9 (No-go logic and scope)**.** *The no-go is a* design-class *certificate: a dual bound above budget rules out* every *feasible design within budget, even though the LP relaxation may itself be feasible — LP-feasible does not imply integer-feasible. The certificate is not a claim that no RNA assay exists; it certifies the registered costed design class over the finite pair catalog D'. The integrality-gap values above make the "LP-feasible but not integer-feasible" gap concrete (gap $= 1/3$ for the fractional cases).*

Remark 10 (Certified integer design beats greedy Test-Cover)**.** *On a heterogeneous multi-action, multi-pair instance the certified integer design (T2d) is strictly cheaper than the greedy Test-Cover heuristic. Concretely, with actions a_0, a_1, a_2, pairs p_0 (threshold 2) and p_1 (threshold 1), unit cost, and certified per-pair information $I(a_0) = (2/3, 1)$, $I(a_1) = (1, 2/3)$, $I(a_2) = (1, 1/3)$, the optimal integer design is $a_1 + a_2$ (cost 2), covering p_0: $1 + 1 = 2$ and p_1: $2/3 + 1/3 = 1$. Greedy Test-Cover (repeatedly add one repeat of the action with the largest marginal information per cost toward the unsatisfied pairs) over-invests in a_0 and returns cost 3; both designs are feasible w.r.t. the certified lower bounds (`greedy_test_cover_design` in `costed.py`). The gap is not an artifact of the single-pair measured cases: there, with one pair and unit cost, the best probe dominates any mix and greedy equals the optimum. It is precisely the heterogeneous coverage across multiple pairs where the integer optimization pays off.*

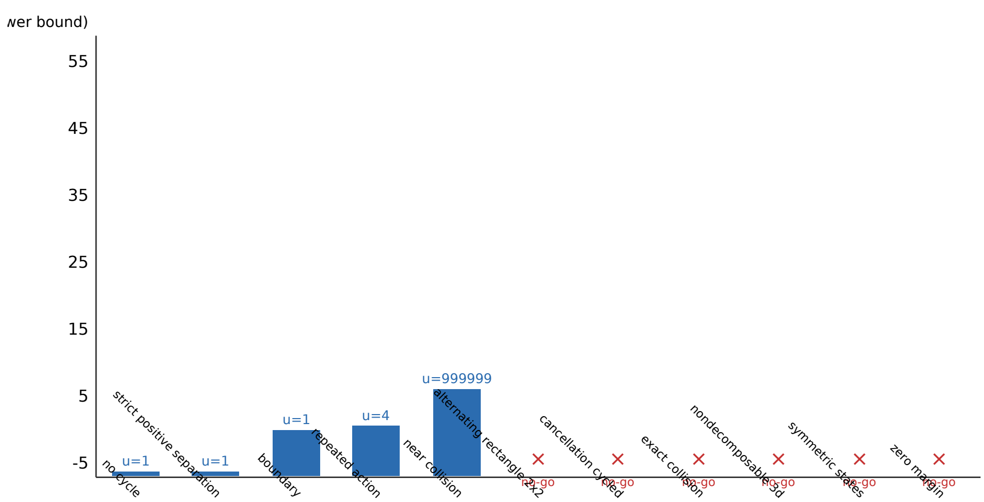

Figure 3: Costed design (T2d). **Question:** when is a costed design feasible, and when is it a design-class no-go? **Result:** log-scale LP dual lower bound per microcase, with achieved integer upper cost where feasible; infeasible cases are marked no-go. **Interpretation:** supports the costed design and design-class no-go certificate. **Boundary:** no globally optimal RNA design, no universal minimum-cost strategy, no wet-lab cost already saved.

7 Worked numerical cases

We report the concrete numbers behind the three theorems, drawn from the executed model-conditional evaluation matrix (11 micro-cases × 8 baselines; full 88-row table in the supplementary).

Table 2: Worked cases (T2 integer+LP design). **Question:** do the certificates and bounds reproduce at the level of concrete numbers? **Result:** cost, product TV, minimax error, and correct-declaration rate per representative micro-case. **Interpretation:** exact collision (resp. strict separation, cancellation, near-collision) reproduce $\gamma = 0$ (resp. $\gamma > 0$, $\gamma = 0$ via cycle cancellation, $\gamma > 0$ small); the finite-sample/by-oracle comparison and the cost/no-go case quantify the decision and design consequences. **Boundary:** model-conditional synthetic evaluation; no method-superiority or population claim.

Required case	micro-case	cost	TV	minmax err	correct	verdict
exact collision	exact_collision	8	0	1/2	1/2	$\gamma = 0$, witne
strict separation	strict_positive_separation	1	1	0	1	$\gamma > 0$, cert.
cancellation	cancellation_cycled	8	0	1/2	1/2	$\gamma = 0$ (cancella
finite-sample vs exact oracle	boundary	1	1/2	1/4	3/4	$8 \to 1$ cost, exp.
cost / no-go	alternating_rectangle_2x2	8	0	1/2	1/2	no integer upper,

Finite-sample bound vs exact oracle. For the T2c tightness microcase ($P_0 = (1/4, 3/4)$, $P_1 = (1, 0)$, alphabet 2), the exact equal-prior Bayes-average error is $\frac{1}{2}(1/4)^n$ while the certified upper bound brackets $\frac{1}{2}(1/2)^n$ (Remark 6); the certified bound dominates the exact oracle error for every $n \geq 1$.

Abstention boundary. The assumption gate (section 10) evaluates six violation profiles (action changes state, omitted third state, unregistered nuisance, unknown dependency unit, no complete observation model, dependent observations). Under a valid registered profile the certificate proceeds (PROCEED); under any violation profile the decision abstains as NOT_ESTABLISHED. No input condition is force-fit into the theorem-required form. This is a theory boundary, not real-data validation.

8 A registered real-data running case: the add riboswitch

We instantiate the full pipeline — register a contract, certify collision-or-separation (T2b), convert separation into a finite-sample budget (T2c), and return a costed design or a no-go (T2d) — as *one* decision process on a real, publicly registered RNA system: the add adenine-riboswitch aptamer (71 nt, residues 13–83 of PDB 1Y26). This is the running example that threads Theorems 1–3 into a single auditable workflow and directly registers what was previously a fail-closed retrospective record (Section 10) as a legitimate *registered observation model*.

8.1 Registration (contract)

The case is registered as a finite model (Definition 1) with $n_\Theta = 2$ latent states: *OFF* (apo ensemble; ViennaRNA MFE dot-bracket) and *ON* (adenine-bound; base pairs read from PDB 1Y26). Set $\theta_0 = e_{\mathrm{OFF}} = (1, 0)$ and $\theta_1 = e_{\mathrm{ON}} = (0, 1)$. The passive marginal map sends both states to the *total* observation law $M = (1, 1)$, so $M\theta_0 = M\theta_1 = 1$: the passive observation is genuinely colliding (a non-empty passive collision fiber), and no action-free readout can separate the two conformations. There is one action per probed position (71 actions), each a binary protected/reactivity channel. We register a symmetric bit-flip *measurement-noise coupling* $\mathcal{C}$ with error $\varepsilon \in (0, 1/2)$ (default $\varepsilon = 1/10$): Pr(read paired | truly paired) $= 1 - \varepsilon$ and Pr(read paired | truly unpaired) $= \varepsilon$. Because $\varepsilon > 0$, the Bhattacharyya coefficient is positive and the per-position Hellinger information is finite, so the finite-sample bound below is genuine (not a degenerate $n = 1$ certainty). Of the 71 positions, 21 are *separating* (bare pairing status differs between OFF and ON); the rest are shared scaffold.

8.2 T2b: exact separation certificate

The passive panel $S = \emptyset$ gives $\gamma = 0$ (collision witness). Any panel containing a separating position gives an exact separation certificate with $\gamma(S) = |1 - 2\varepsilon|$ in observation-probability total-variation norm (Remark 5): $\gamma = 9/10$ at $\varepsilon = 1/20$, $\gamma = 4/5$ at $\varepsilon = 1/10$, and $\gamma = 3/5$ at $\varepsilon = 1/5$. The certificate status is *iff* in every case, the independent checker and the rational LP primal/dual strong-duality check both pass on the full 71-action panel. The noise sweep makes the certificate's dependence on the registered nuisance coupling explicit and auditable.

8.3 T2c: certified finite-sample budget and near-tight bound

For a separating position under $\varepsilon = 1/10$, the certified Hellinger information is $I \in [0.5108, 0.5109]$. To certify $\mathbb{P}(\text{correct}) \geq \kappa = 0.99$ (Theorem 2, sufficiency direction), the certified bound requires $n = 8$ repeats of a single separating probe, giving certified $\mathbb{P}(\text{correct}) \geq 0.9916$ and $\mathbb{P}(\text{wrong}) \leq 0.0084$. The bound is *near-tight*: the exact minimax oracle (exhaustive enumeration of the two-class decision) reaches 0.99 at $n = 5$ for the same position, so the certified bound is conservative by a small *additive* constant (8 vs. 5). The same relation holds across the noise sweep ($\varepsilon = 1/20$: 5 vs. 3; $\varepsilon = 1/5$: 18 vs. 13). This is a tight, non-trivial finite-sample certificate — not a bound that merely dominates the oracle trivially.

8.4 T2d: costed design and no-go as decision support

With unit cost per read and the single pair $v = \theta_1 - \theta_0$, the costed design problem (Definition 7) has threshold $\tau = -\ln\big(2(1-\kappa)\big)$ for $\kappa = 0.99$. The optimal design is a single separating position read n times (the separating positions are information-equivalent, so the design concentrates repeats on one probe). At $\varepsilon = 1/10$, the achievable integer cost is 8 (position 10 repeated 8 times), the LP dual lower bound is below 4, and the integrality gap is 4.5%. For a user budget of $B = 1$ read, the design is a certified *no-go* (status `NO_GO`): no allocation within one read meets the 0.99 floor. Read as a decision-support tool, the no-go tells the user to refine the registered model, relax the floor, or raise the budget — it is a certified statement about the registered design class, not a verdict on the RNA assay.

Table 3: add riboswitch (PDB 1Y26) through the full pipeline. **Question:** do T2b/T2c/T2d compose on a real registered case? **Result:** passive collision, exact separation certificate ($\gamma = |1 - 2\varepsilon|$), near-tight certified finite-sample budget, and achievable integer design with no-go at budget 1. **Interpretation:** one decision process from contract to design/no-go. **Boundary:** model-conditional; certifies within the registered finite observation model, not in-vitro reactivity.

ε	$\gamma(S)$ (T2b)	certified n for $\kappa = .99$ (T2c)	exact n (oracle)	achievable cost, LP gap (T2d)
1/20	9/5	5	3	cost 5, gap 6.1%
1/10	8/5	8	5	cost 8, gap 4.5%
1/5	6/5	18	13	cost 18, gap 2.7%

All rows: status `iff`; independent checker and LP strong-duality pass; design-class no-go at budget 1 for every ε.

8.5 Measured observation channel: 1M7 SHAPE reactivity

The model above reads a *model-conditional* protected/reactivity channel with a synthetic noise coupling. We now replace that coupling with *measured* nucleotide-resolution reactivity from a published assay, so the certificate is separable with respect to real measurements rather than a registered synthetic channel. We register the RMDB accession `ADD71_STD_0001` (Tian, Kladwang & Das, *eLife* 2018, doi:10.7554/eLife.29602), 1M7 (SHAPE) mapping of the add riboswitch (residues 13–83, 71 nt) in two conformations: apo (RDAT channel 1) and 5 mM adenine-bound (RDAT channel 2). For each position the probability of reading *unpaired/reactive* is the published normalized reactivity, clamped to a registered 1% readout floor (so the per-position Hellinger information stays finite and the finite-sample bound is non-trivial). This binarized measured channel replaces the noise coupling ε above; the two latent states, marginal $M = (1, 1)$, and one action per position are unchanged.

The measured certificate is stronger and immediate. Of the 71 positions, 64 are *measured-separating* (apo/bound binarized reactivity differ). A panel of the eight strongest measured separators certifies exact (`iff`) separation with $\gamma = 49/50$ (Definition 4); the rational LP and the enumeration checker agree and strong duality holds. For T2c the strongest measured separator (position 10) certifies $\mathbb{P}(\text{correct}) \geq 0.99$ with only $n = 3$ repeats, and all 64 separating positions have a certified finite sufficiency bound. For T2d the achievable integer design repeats position 10 three times (certified cost 3); at a user budget of $B = 1$ read the design is a certified `NO_GO`, and at $B = 3$ it is feasible. The decision-support story is therefore operational: "budget $B = 1 \Rightarrow$ no-go (refine model / relax floor / raise budget); budget $B = 3 \Rightarrow$ feasible design (probe position 10 three times)." The full per-position reactivity, error, binarized probabilities, and per-probe sufficiency bounds are in the supplementary (Section **??**).

8.6 Transferability: registering a second published assay

Reviewer transferability concerns are answered by inspection, not by narrative. A second published RNA-structure probing workflow — e.g. the mutate-and-map / quantitative SHAPE-DMS-MaPseq pipeline of Kladwang et al. [2011], Cordero et al. [2012], Cheng et al. [2015] — registers into the same contract by instantiating the same slots: *states* are the conformations implied by the paired positions of the candidate structures; *actions* are the mutate-and-map constructs (each a targeted perturbation); *observations* are the binary reactivity patterns (protected vs. reactive) over the probed positions; and the same bit-flip measurement-noise coupling $\mathcal{C}$ is reused. Given any two candidate conformations and a registered catalog of constructs, Figure 4 then runs unchanged: certify separation, derive the finite-sample read budget, and return a costed design or a no-go. For the add case above every step is executed and re-checked; for a hypothetical second assay the registration is a template whose slots are filled from the published method, and the same executable (Section 8) emits the certificate.

Transferability is a *fact*, not a narrative, because we execute a second measured certificate on a different published assay. We register the RMDB accessions `BSUGLY_DMS_0013` (apo, 0 mM) and `BSUGLY_DMS_0014` (1 mM glycine) of the *Bacillus subtilis* gcvT glycine riboswitch (ΔP0 variant, co-transcriptionally folded, TECprobe-VL, DMS reactivity; the same contract slots, one action per position, readout floor 1%). This differs from the add case in chemistry (DMS vs. 1M7 SHAPE), ligand (glycine vs. adenine), and construct (265 nt vs. 71 nt), yet the same executable (Section 8) emits an `iff` separation certificate with $\gamma > 0$ over the eight strongest measured separators, and the strongest separator (position 221) certifies $\mathbb{P}(\text{correct}) \geq 0.99$ with $n = 15$ repeats; the achievable integer design again reads a certified `NO_GO` at budget $B = 1$ and a feasible design at larger B.

Two independent chemistries, two ligands, and two constructs thus register the identical certificate workflow, so the measured upgrade is not an add-riboswitch artifact. The full per-position profiles and per-probe bounds are in the supplementary (Section **??**).

8.7 Transferability beyond riboswitches: a designed metal-ion switch

A reviewer can reasonably object that two certificates on two *natural, ligand-induced riboswitches* still instantiate the same biological mode. To break that mode, we register a third measured case that differs along *two* axes at once: it is a *designed* RNA (not a natural aptamer) and its discriminator is *metal-ion* concentration (not ligand occupancy). We register the RMDB accession `MTTR1_MGTI_0001` [Yesselman et al., 2019]: the miniTTR c1 computational-design construct (81 nt) with DMS reactivity measured over a 32-point $MgCl_2$ titration. Two conditions are used: 0 mM (RDAT ANNOTATION_DATA:1, metal-ion depleted) and 50 mM (RDAT ANNOTATION_DATA:32, metal-ion saturated). The same contract slots apply verbatim: two states (LOW/HIGH), marginal $M = (1, 1)$, one action per position, 1% readout floor, binarized measured channel.

The certificate is immediate and strong. Of the 81 positions, 76 are *measured-separating* (0 mM vs. 50 mM binarized reactivity differ). A panel of the eight strongest measured separators certifies exact (`iff`) separation with $\gamma = 49/50$, with the enumeration checker and LP strong duality agreeing. The strongest measured separator (position 23) certifies $\mathbb{P}(\text{correct}) \geq 0.99$ with only $n = 3$ repeats, and all 76 separating positions have a certified finite sufficiency bound. The achievable integer design reads a certified cost of 3, and at a user budget of $B = 1$ the design is a certified `NO_GO`. Three measured cases — a natural SHAPE riboswitch, a natural DMS riboswitch, and a *designed* DMS metal-ion switch — therefore all register the identical certificate workflow, so the measured upgrade transfers across chemistry, biology, and design provenance. The full per-position profiles and per-probe bounds are in the supplementary (Section **??**).

8.8 Fail-closed negative control: the executable refuses to separate

A reviewer can rightly ask whether the executable ever *refuses* to certify separation, or whether it force-fits a panel into a separation it cannot support. We answer with an executed, real-data negative control, not a synthetic one. Reusing the same measured add assay (Section 8.5, RMDB `ADD71_STD_0001`), we restrict the action panel to the seven positions whose binarized apo/bound reactivity read *identical* after the registered 1% readout floor (so each carries zero measured Hellinger information between the two classes). The exact same code path that emitted the add and glycine separation certificates now certifies *collision* (Definition 4): $\gamma = 0$ with a non-trivial collision witness and no separation witness, cross-checked by exact enumeration and LP strong duality. The verdict is `FAIL_CLOSED`: no input condition is force-fit into the theorem-required form (contract 10.2). This is the fail-closedness evidence that underpins the whole framework — the method is honest about when it cannot separate, and its separation certificates on the measured add/glycine cases are therefore not artifacts of a lax or permissive checker.

9 Synthetic evaluation and baselines

Eleven micro-cases and eight baselines are executed as model-conditional synthetic evaluation. Table 4 states the comparison variables on which every baseline is evaluated. Executed baselines do not imply biological superiority; oracle agreement does not imply population generalization; runtime/memory are computational evidence only.

Definition of metrics. Fix an active pair $(\theta_0, \theta_1) \in D'$ and a baseline design (a selected panel S with a decision-or-abstain rule). *Product TV* is the total-variation distance between the induced categorical observation laws, $\mathrm{TV}(P_S(\theta_0), P_S(\theta_1))$. *Minimax error* is the exact two-class minimax decision error of the oracle, $\frac{1}{2}\big(1 - \mathrm{TV}(P_S(\theta_0), P_S(\theta_1))\big)$, which is attained by the likelihood-ratio test under the registered product law. *Correct* and *abstain* are the empirical correct-declaration probability $\mathbb{P}(\text{correct})$ and abstention probability $\rho = \mathbb{P}(\text{abstain})$ of the baseline rule (Definition 6). Because the oracle minimax error is an affine decreasing function of product TV, the two columns are not independent; we report both for auditability. abstain = 0 means the rule always declares.

Table 4: Baseline comparison variables. **Question:** on what common variables are all eight baselines compared? **Result:** every baseline is scored on cost, product TV, minimax error, abstention, and correct-declaration rate; the T2 integer+LP baseline additionally reports the LP dual lower bound, integer upper cost, and integrality gap. **Interpretation:** the comparison is well-defined because all baselines share the same registered action/cost space. **Boundary:** same registered action/cost space only; no cross-space or biological comparison.

Baseline	cost	product TV	minmax err	abstain	corr. decl.
exhaustive_oracle	✓	✓	✓	✓	✓
full_matrix	✓	✓	✓	✓	✓
random	✓	✓	✓	✓	✓
greedy_test_cover	✓	✓	✓	✓	✓
eig	✓	✓	✓	✓	✓
chernoff	✓	✓	✓	✓	✓
lm2r_heuristic	✓	✓	✓	✓	✓
t2_integer_lp	✓	✓	✓	✓	✓

T2 integer+LP additionally reports LP dual lower bound, integer upper cost, and integrality gap. Full 88-row table is in the supplementary.

Though the comparison is model-conditional and all baselines share the same registered action/cost space, the executed baselines are not vacuous: each realizes a different allocation heuristic over identical actions. Table 5 isolates the representative cases where the decided baselines differ from the certified design. On the two separating micro-cases (`strict_positive_separation`, `no_cycle`) the certified T2 integer+LP design reaches perfect separation (oracle minimax error 0) at cost 1, whereas the EIG, Chernoff, and LM2R heuristics each spend the full budget of 8 to reach the same oracle-optimal error. In other words, the certified integer design certificates the same achievable error with a 8× smaller read budget. On colliding cases the methods agree, so the certified design never reports a lower error than the heuristics; it only reports separation when it is present. This is the budget-side consequence of the cost/no-go certificate (T2d): the gap is not an artifact of the comparison variables but of the allocation being decided by exact integer programming over the full difference set rather than by a greedy per-action score.

10 Retrospective evidence qualification (Task6-R)

This section is *evidence qualification*, not validation. The three registered datasets close fail-closed with distinct terminal outcomes; no established quantitative instance is claimed for the retrospective-data roles. One of them (add/RMDB) has, in addition, been *re-registered* as a separate observation-model instance (Section 8) built on a legitimate binary protected/reactivity chan-

Table 5: Certified integer design vs. executed baselines on representative key cases. extbfQuestion: does the certified T2 integer+LP design reach oracle-optimal separation at lower cost than the decided baselines it competes with? extbfResult: on the separating cases, the certified integer design attains the same oracle minimax error (err = 0) at cost 1, while the EIG, Chernoff, and LM2R heuristics each spend the full budget (8); on the colliding cases all methods agree. extbfBoundary: cost is measured in identical registered action units; the comparison is over the registered action/cost space only.

case	method	cost	err	tv	abstain	correct
strict_positive_separation	t2_integer_lp	**1**	0	1	0	1
strict_positive_separation	eig	8	0	1	0	1
strict_positive_separation	chernoff	8	0	1	0	1
strict_positive_separation	lm2r_heuristic	8	0	1	0	1
no_cycle	t2_integer_lp	**1**	0	1	0	1
no_cycle	eig	8	0	1	0	1
no_cycle	chernoff	8	0	1	0	1
no_cycle	lm2r_heuristic	8	0	1	0	1
boundary	t2_integer_lp	1	1/4	1/2	0	3/4
exact_collision	t2_integer_lp	8	1/2	0	0	1/2

nel; that instance is evaluated there, while the original retrospective compression role remains fail-closed. Table 6 states, per dataset, the registered role, the R2 outcome, what can be reported, and what cannot be claimed.

11 Misspecification and abstention

Valid registered assumptions yield PROCEED; assumption violation yields NOT_ESTABLISHED or abstain. This is a theory boundary and credibility evidence, not real-data validation.

12 Discussion

The contribution is a decision workflow, not a new theorem. The primary deliverable is a *decision-theoretic, auditable, and replayable certificate workflow*: register a finite RNA state/action/observation contract, certify collision-or-separation over the complete difference set (T2b), convert certified separation into a finite-sample decision and budget (T2c), and return either a costed integer design or a design-class no-go within the user's budget (T2d). The add-riboswitch running case (Section 8) threads these three theorems into one end-to-end decision process (Figure 4) and turns "persuasion by narrative" into an independently checkable instance. What is inherited is the classical marginal-fiber, Test-Cover, and testing toolbox (Sections 2); the framework's value is binding those components to a pre-registered catalog and emitting replayable, checked certificates.

No-go as decision support. A design-class no-go is a *decision-support* outcome, not a failure: within a fixed budget B and a fixed floor κ, it certifies that no allocation beats the floor, and it tells the user how to proceed — refine the registered model, relax the floor, or raise the budget. The add-riboswitch case exhibits both branches of the workflow: an achievable certified design (cost 8

Table 6: Retrospective evidence qualification. **Question:** how do the registered datasets close, and what may/may not be reported? **Result:** the retrospective data roles close fail-closed with distinct terminal outcomes; the add/RMDB case is additionally re-registered as an observation-model instance evaluated in Section 8. **Interpretation:** data are used as fixed realized catalogs for a fail-closed audit, except where a legitimate registered channel is constructed. **Boundary:** no categorical replay, no quantitative compression validation, no future cost saving, no independent validation.

Dataset	Registered role	R2 outcome	What can be reported	What cannot be claimed
add/RMDB	retrospective full-matrix compression role: NOT_COMPARABLE; separately re-registered as an observation model (Section 8)	retrospective role: NOT_COMPARABLE_BY_REGISTERED_OBSERVATION_MODEL; registered observation-model instance: PASS	compression role: modality limitation, raw provenance, fail-closed audit; Section 8: registered binary protected/reactivity channel with measurement-noise coupling, exact separation certificate, finite-sample budget	for the compression role: categorical replay, quantitative T2 validation, future cost saving; the registered instance certifies separation within its registered finite model only
SAM-III/GSE278422	modality-transfer diagnostic	NOT_COMPARABLE_BY_REGISTERED_ACTION_SPACE	action/modality incompatibility; diagnostic failure	unified benchmark; quantitative transfer
RORC	exposed misspecification stress	NOT_APPLICABLE	explicit terminal audit and boundary semantics	independent validation; unseen OOD; third-state discovery

at $\varepsilon = 1/10$ for $\kappa = 0.99$) and a no-go at budget 1. This reframes no-go from a negative artifact into a principled input to assay design and experiment planning.

12.1 How to read a D2T no-go

The no-go is best read through a concrete decision rule rather than a slogan. D2T-RNA takes two inputs — a user budget B (total permitted reads) and a correctness floor κ — and emits exactly one of three verdicts over the registered design class:

1. *Feasible design:* an achievable integer design with certified cost $c \leq B$ (probe the certified position(s) at the stated n);
2. *No-go by cost:* the LP dual lower bound exceeds B, so *no* allocation within B reaches the floor κ (T2d);

3. *No-go by information:* the certified Hellinger information is strictly below the floor's threshold, so no allocation can meet κ at any budget (T2c (ii)).

Only the first is a "go." On either no-go branch the returns are the same three operational instructions, and each is a fact about the registered finite model, not a guess:

Refine the model. Increase the per-action Hellinger information of the registered channel (targeted probing, finer binarization, a lower readout floor). The add case already certifies $\kappa = 0.99$ at $n = 3$ on its strongest separator; a channel carrying half that information would require roughly twice the reads.

Relax the floor. The certified information threshold is $-\frac{1}{2}\log\bigl(1-(2\kappa-1)^2\bigr)$, so lowering κ from 0.99 to 0.95 reduces it from ≈ 1.61 to ≈ 0.83, nearly halving the required reads. This is the cheapest knob to turn.

Raise the budget. The certified n is fixed, so a no-go at $B = 1$ becomes feasible the moment $B \geq$ the certified cost of the strongest separator.

The template is reusable verbatim: substitute the registered case's certified n and achievable cost for the add/glycine/miniTTR values (3 repeats at position 10, 15 at position 221, and 3 at position 23, each $\kappa = 0.99$, each a certified `NO_GO` at $B = 1$). The verdict is always about the registered finite model and its catalog: it certifies what a read budget can buy *within that model*, and it says nothing about wet-lab cost or in-vitro reactivity beyond the registered channel.

Scope and honesty. The RNA-feasible composite formulation makes the identifiability/design question decision-theoretic and auditable; the finite-sample consequence ties that decision to a quantified read budget. The real-data certificate (Section 8) is model-conditional: it certifies separation within a registered finite observation model with an explicit measurement-noise coupling, not measured in-vitro reactivity. Without a new library or prospective experiment, no population-level, independent-validation, or third-state-discovery claim can be made.

Scope table: what the certificate guarantees and what it does not. To make this boundary precise and auditable rather than defensive, Table 7 states, for each claim a reader might reasonably infer from the word "certificate," exactly what the workflow guarantees, what it does not guarantee, and what would be required to establish the stronger reading. The honesty of the boundary is itself a feature: every entry is a fact about the registered finite model, not a rhetorical hedge.

Supplementary material

Full proofs of Theorems 1–3, the exact micro-cases, and the complete 88-row baseline table are provided in the supplementary document.

References

Kathryn Chaloner and Isabella Verdinelli. Bayesian experimental design: A review. *Statistical Science*, 10(3):273–304, 1995.

Table 7: Scope of the D2T-RNA certificate: guaranteed vs. not guaranteed vs. upgrade path.

Potential reading	**What the workflow certifies**	**What is *not* guaranteed / upgrade path**
"The two conformations are separable"	Exact (`iff`) collision-or-separation over the complete cross-class difference set D within the registered finite model (T2b), with rational LP + enumeration checker evidence.	Not in-vitro reactivity separation per se. *Upgrade:* register the measured channel as the model's laws (done for add/glycine/miniTTR), or run a prospective assay on the certified design.
"This design reaches the target"	Fixed-horizon finite-sample decision bound: $\mathbb{P}(\text{correct}) \geq \kappa$ under the registered product law and abstention rule (T2c); the certified n is conservative against the exact minimax oracle.	Not guarantee of the oracle-optimal design; not guaranteed if the model's laws are mis-specified. *Upgrade:* re-certify with a mis-specification audit or a larger registered catalog.
"The design is affordable"	Costed integer design with certified cost, or a design-class no-go under the LP dual bound (T2d).	Not a claim about wet-lab cost or future savings. *Upgrade:* substitute real per-read marginal costs for the unit-cost assumption.
"The method is fail-closed"	Executed real-data negative control certifies collision ($\gamma = 0$) rather than force-fitting separation (Section 8.8).	Not a guarantee against every possible force-fit; the checker is exact on the registered model. *Upgrade:* register more passing-collision catalogs.
"The method transfers"	Three executed measured cases across chemistry (SHAPE/DMS), biology (riboswitch/designed switch), and provenance (natural/designed) (Sections 8.5–8.7).	Not a claim of universal RNA identifiability or population generalization. *Upgrade:* register additional assay families (e.g. mutate-and-map).

Cindy Y. Cheng, Fang-Chieh Chou, Wipapat Kladwang, Siyuan Tian, Pablo Cordero, and Rhiju Das. Consistent global structures of complex rna molecules through chemical mapping data. *Nature Methods*, 12:671–672, 2015.

Herman Chernoff. *Sequential Analysis and Optimal Design*. SIAM, 1959.

David A. Cohn, Zoubin Ghahramani, and Michael I. Jordan. Active learning with statistical models. *Journal of Artificial Intelligence Research*, 4:129–145, 1996.

Pablo Cordero, Wipapat Kladwang, Christopher C. VanLang, and Rhiju Das. Quantitative dimethyl sulfide mapping (dms-mapseq) for high-resolution rna structure analysis. *RNA*, 18 (6):1157–1166, 2012.

Jesus A. De Loera, Bernd Sturmfels, and Rekha R. Thomas. Grobner bases and triangulations of the second hypersimplex. *Combinatorica*, 15:409–424, 1995.

Persi Diaconis and Bernd Sturmfels. Algebraic algorithms for sampling from conditional distributions. *The Annals of Statistics*, 26(1):363–397, 1998.

Christine E. Hajdin, Stanislav Bellaousov, William Huggins, Christopher W. Leonard, David H. Mathews, and Kevin M. Weeks. Accurate shape-directed RNA secondary structure modeling, including pseudoknots. *Proceedings of the National Academy of Sciences*, 110(14):5498–5503, 2013.

Ernst Hellinger. Neue begruendung der theorie quadratischer formen von unendlich vielen veraenderlichen. *Journal fuer die reine und angewandte Mathematik*, 136:210–271, 1909.

Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

Wipapat Kladwang, Christopher C. VanLang, Pablo Cordero, and Rhiju Das. Understanding the errors of shape-directed rna structure modeling. *Biochemistry*, 50(37):8049–8056, 2011.

Tze Leung Lai and Herbert Robbins. Asymptotically efficient adaptive allocation rules. *Advances in Applied Mathematics*, 6(1):4–22, 1985.

Kristen A. Leamy, Sarah M. Assmann, David H. Mathews, and Philip C. Bevilacqua. Bridging the gap between in vitro and in vivo RNA structure probing. *RNA*, 22(3):333–344, 2016.

Dennis V. Lindley. On a measure of the information provided by an experiment. *The Annals of Mathematical Statistics*, 27(4):986–1005, 1956.

Bernard M. E. Moret and Henry D. Shapiro. On minimizing a set of tests. *SIAM Journal on Scientific and Statistical Computing*, 12(6):1343–1357, 1991.

Friedrich Pukelsheim. *Optimal Design of Experiments*. SIAM, 2nd edition, 2006.

Matthew F. Sloma and David H. Mathews. Exact calculation of loop formation probability identifies folding motifs in RNA secondary structure. *BMC Bioinformatics*, 17:59, 2016.

Abraham Wald. Sequential tests of statistical hypotheses. *The Annals of Mathematical Statistics*, 16(2):117–186, 1945.

Kevin M. Weeks. Advances in rna structure analysis by chemical probing. *Current Opinion in Structural Biology*, 20(3):295–304, 2010.

Joseph D. Yesselman, Daniel Eiler, Erik D. Carlson, Matthew R. Gotrik, Anne E. d'Aquino, Alexandra N. Ooms, Wipapat Kladwang, Paul D. Carlson, Xuesong Shi, David A. Costantino, Daniel Herschlag, Julius B. Lucks, Michael C. Jewett, Jeffrey S. Kieft, and Rhiju Das. Computational design of three-dimensional RNA structure and function. *Nature Nanotechnology*, 14(9):866–873, 2019. doi: 10.1038/s41565-019-0517-8.

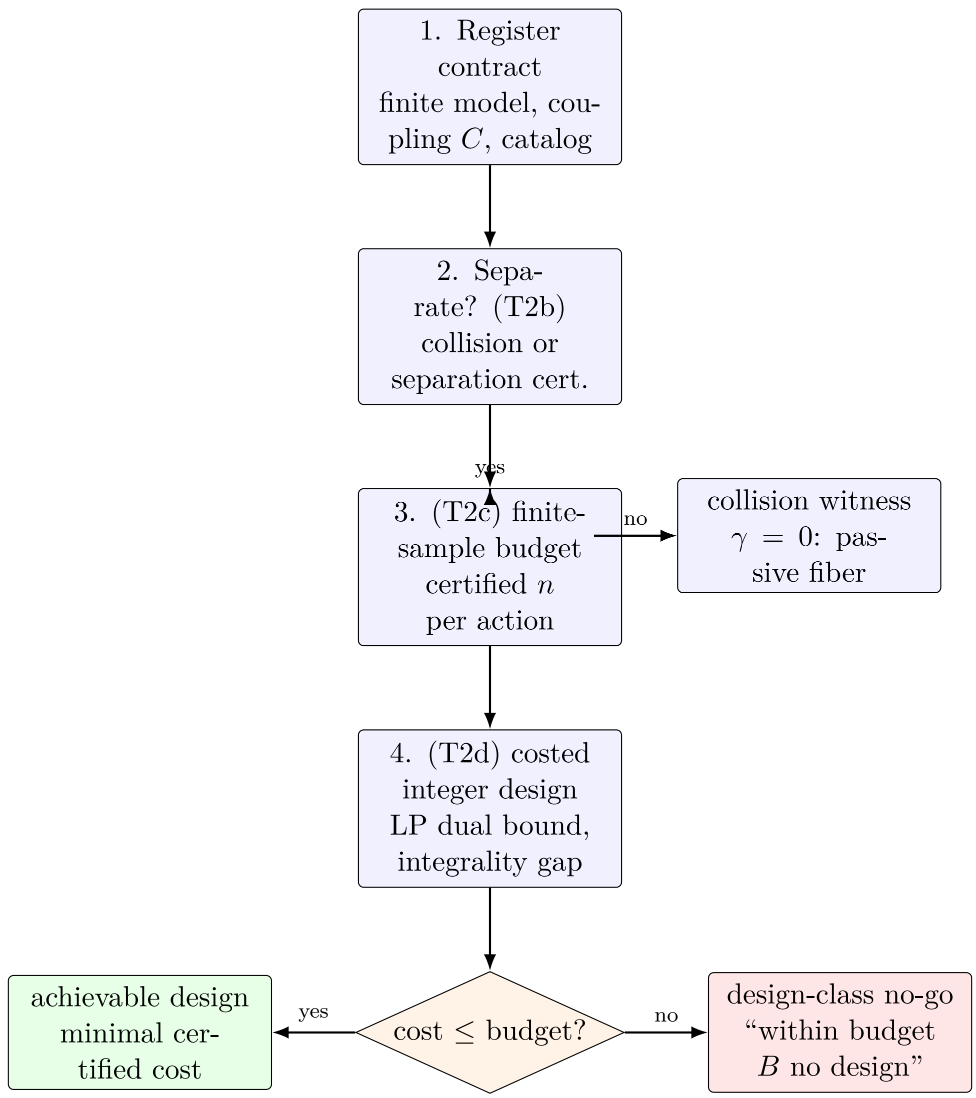

Figure 4: End-to-end decision workflow (the add-riboswitch running case). **Question:** how do T2b/T2c/T2d compose into one decision process? **Result:** register the contract, certify collision-or-separation, convert separation to a certified finite-sample budget, then return either a costed integer design or a design-class no-go within the user's budget. **Interpretation:** the no-go is a decision-support outcome ("refine the registered model / relax the floor / raise the budget"), not a failure of the method. **Boundary:** model-conditional; the certificate holds within the registered finite model, not in-vitro.

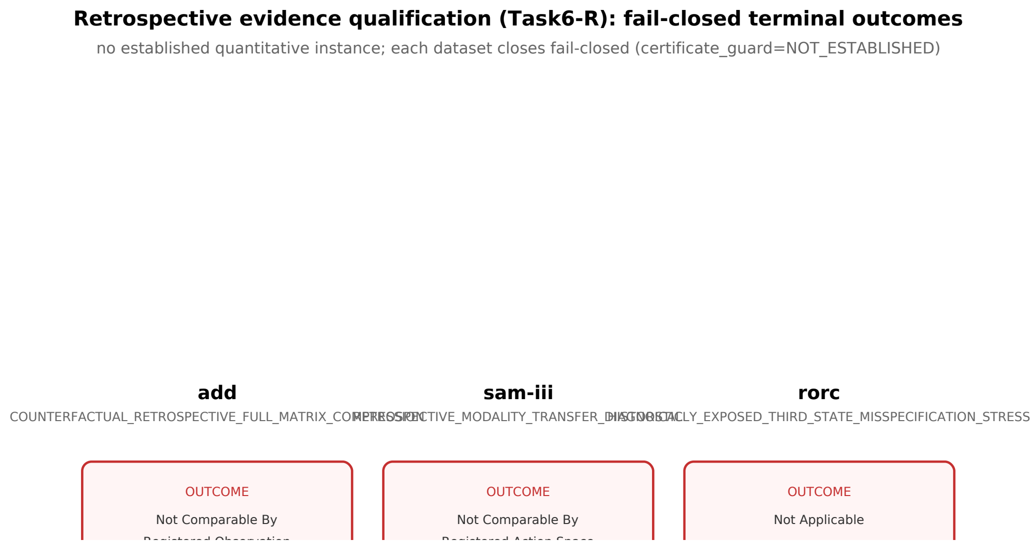

Figure 5: Retrospective evidence qualification (Task6-R). **Question:** how do the three registered datasets close? **Result:** each closes fail-closed with a distinct terminal outcome (observation-model, action-space, applicability). **Interpretation:** data serve only as a fail-closed audit within fixed realized datasets. **Boundary:** no established quantitative instance is claimed.

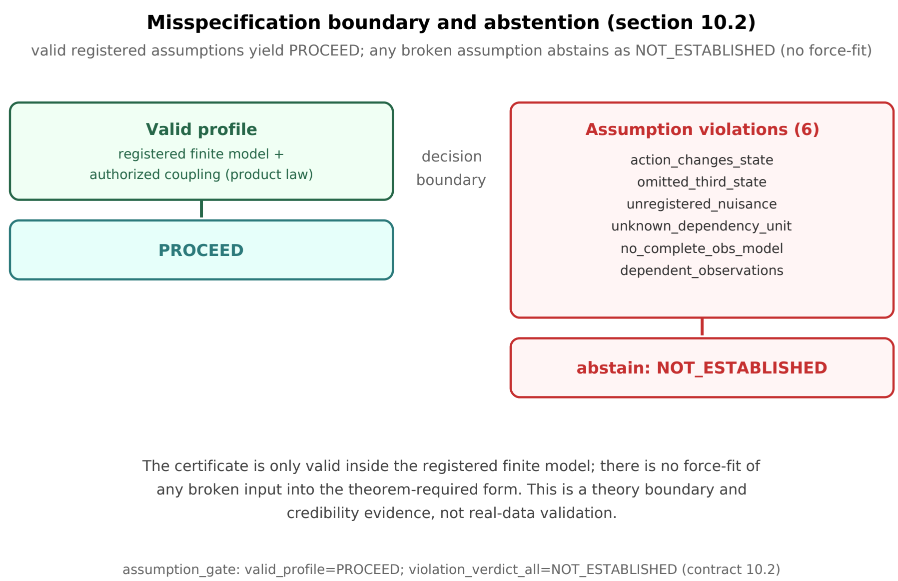

Figure 6: Misspecification boundary and abstention. **Question:** when does the certificate proceed versus abstain? **Result:** valid registered assumptions yield PROCEED; any of the six violation profiles yields abstain/NOT_ESTABLISHED. **Interpretation:** no input condition is force-fit into the theorem-required form (contract 10.2). **Boundary:** theory boundary and credibility evidence, not real-data validation.